

AMAT113 Calculus II

Homework 5

Sequences and Infinite Series

Submission Instructions

- Homework is due Sunday at 11:59 PM Eastern Time.
- Please print this PDF and complete the problems neatly by hand whenever possible.
- If printing is not available, completing the work on loose-leaf paper is acceptable.
- Show all necessary work. Scan and submit one PDF through Brightspace.
- Verify all pages are included, upright, and legible.

Part I: Sequences

Properties of Limits of Sequences: If $\lim_{n \rightarrow \infty} a_n = A$, $\lim_{n \rightarrow \infty} b_n = B$, and c is any real number, then

1. $\lim_{n \rightarrow \infty} (a_n + b_n) = A + B$
2. $\lim_{n \rightarrow \infty} (ca_n) = cA$
3. $\lim_{n \rightarrow \infty} (a_n b_n) = AB$
4. $\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = \frac{A}{B}, \quad b_n \neq 0, \quad B \neq 0.$

1. Determine whether the following sequences converge or diverge. If the sequence converges, then compute its limit.

(a) $\left\{ \frac{6n^3 + 5n^2 + 7}{4n^3 - 2n + 2} \right\}$

(b) $\left\{ \ln(n+4) - \frac{1}{2} \ln(n) \right\}$

(c) $\left\{ (-1)^n \left(\frac{n+1}{n} \right) \right\}$

Part I: Sequences (continued)

$$(d) \left\{ \frac{\sqrt{2n^2 + 1}}{n} \right\}$$

$$(e) \left\{ \frac{e^n - e^{-n}}{e^n + e^{-n}} \right\}$$

$$(f) \{ \sqrt{n+1} - \sqrt{n} \}$$

Part I: Sequences (continued)

Suppose that $\lim_{n \rightarrow \infty} a_n = A$ and each number a_n lies in the domain of the function f . If f is continuous at $x = A$, then $\lim_{n \rightarrow \infty} f(a_n) = f(A)$.

2. Compute the following limits.

(a) $\lim_{n \rightarrow \infty} \tan \left(\frac{\pi n^2 + 1}{3 - 4n^2} \right).$

(b) $\lim_{n \rightarrow \infty} \ln \left(\frac{n^2 + 1}{(n + 2)(n + 3)} \right).$

Part I: Sequences (continued)

Squeeze Theorem for Sequences: Consider sequences $\{a_n\}$, $\{b_n\}$, and $\{c_n\}$. Suppose there is an integer N such that

$$a_n \leq b_n \leq c_n \quad \text{for all } n \geq N.$$

if there is a number L such that

$$\lim_{n \rightarrow \infty} a_n = L = \lim_{n \rightarrow \infty} c_n,$$

then $\{b_n\}$ converges and $\lim_{n \rightarrow \infty} b_n = L$.

3. Determine the limit of the following sequences, if it exists, using the Squeeze Theorem. Justify your answers.

(a) $\left\{ \frac{\cos(n) + \sin(n)}{n + 1} \right\}$

(b) $\left\{ \frac{2^n + 3^n}{4^n} \right\}$

(c) $\left\{ \frac{\sin(n^2)}{n} \right\}$

(d) $\left\{ \frac{(-1)^n + \cos(n)}{n^2 + 1} \right\}$

(e) $\left\{ \frac{n!}{n^n} \right\}$

Part II: Infinite Series

An **infinite series** is an expression of the form

$$\sum_{k=1}^{\infty} a_k = a_1 + a_2 + a_3 + \dots$$

The infinite series

$$\sum_{k=1}^{\infty} a_k$$

is said to **converge** to the **sum** S if

$$S = \lim_{n \rightarrow \infty} S_n,$$

where S_n denotes the n -th **partial sum**

$$S_n = a_1 + a_2 + a_3 + \dots + a_n = \sum_{k=1}^n a_k.$$

If the limit S does not exist, the series $\sum_{k=1}^{\infty} a_k$ is said to **diverge**.

1. Write the given repeating decimal as an infinite series.

(a) $0.66\bar{6}$

(b) $0.92929\bar{2}$

Part II: Infinite Series (continued)

A useful tool to determine if a series diverges is the **Divergence Test**. It states that if $\lim_{n \rightarrow \infty} a_n = c \neq 0$ or $\lim_{n \rightarrow \infty} a_n$ does not exist, then the series $\sum_{k=1}^{\infty} a_k$ diverges.

2. Determine whether the given infinite series converges or diverges. If it converges, find its sum.

(a) $\sum_{k=1}^{\infty} \frac{1}{k(k+1)}$

(b) $\sum_{k=1}^{\infty} \frac{3 \cdot 2^k + 3^k}{5^k}$

(c) $\sum_{k=1}^{\infty} \frac{k}{k+2}$

Part II: Infinite Series (continued)

$$(d) \sum_{k=1}^{\infty} \ln\left(\frac{k}{k+1}\right)$$

$$(e) \sum_{k=1}^{\infty} \cos(\pi k)$$

$$(f) \sum_{k=4}^{\infty} \frac{1}{k^2 - 1}$$

Part III: Divergence and Integral Test

Divergence Test: If $\lim_{n \rightarrow \infty} a_k = c \neq 0$ or $\lim_{n \rightarrow \infty} a_k$ does not exist, then the series $\sum a_k$ diverges.

1. For each of the following series, if the divergence test applies, $\sum a_k$, either state that $\lim_{n \rightarrow \infty} a_k$ does not exist or find $\lim_{n \rightarrow \infty} a_k$. If the divergence test does not apply, state why.

(a) $\sum_{k=1}^{\infty} \frac{k}{\sqrt{3k^2 + 2k + 1}}$

(b) $\sum_{k=1}^{\infty} \frac{2^k + 3^k}{10^k}$

(c) $\sum_{k=1}^{\infty} e^{-2/k}$

(d) $\sum_{k=1}^{\infty} \frac{\ln(k)}{k}$

Part III: Divergence and Integral Test (continued)

The Integral Test: Let a_k for all $k = 1, 2, \dots$ and f be a continuous decreasing function defined on $[1, \infty)$ so that $f(k) = a_k$ for each $k = 1, 2, \dots$. Then

$$\sum a_k \text{ converges} \quad \text{if and only if} \quad \int_1^{\infty} f(x) dx \text{ converges.}$$

That is, the series and the improper integral either both converge or both diverge.

2. Use the Integral Test to determine whether the given infinite series converges or diverges.

(a) $\sum_{k=1}^{\infty} k e^{-k^2}$

(b) $\sum_{k=1}^{\infty} \frac{1}{2k+1}$

Part III: Divergence and Integral Test (continued)

$$(c) \sum_{k=1}^{\infty} \frac{k}{(1+k^2)^3}$$

$$(d) \sum_{k=2}^{\infty} \frac{1}{k \ln(k)}$$

Part IV: Comparison Tests

The Basic Comparison Test: Let $\sum a_k$ and $\sum b_k$ be infinite series with $0 < a_k \leq b_k$ for each $k = 1, 2, \dots$. Then

- (i) If $\sum b_k$ converges, then so does $\sum a_k$.
- (ii) If $\sum a_k$ diverges, then so does $\sum b_k$.

1. Use the Basic Comparison Test to determine whether the given infinite series converges or diverges.

(a) $\sum_{k=1}^{\infty} (k-1)e^{-k}$

(b) $\sum_{k=1}^{\infty} \frac{1}{k^2 + 1}$

(c) $\sum_{k=1}^{\infty} \frac{\sqrt{k}}{k^3 + 1}$

(d) $\sum_{k=1}^{\infty} \frac{1}{k^{1/2} + k^{3/2}}$

Part IV: Comparison Tests (continued)

The Limit Comparison Test: Let $\sum a_k$ and $\sum b_k$ be infinite series with $a_k \geq 0$, $b_k > 0$ for all $k = 1, 2, \dots$. If the limit

$$\rho = \lim_{k \rightarrow \infty} \frac{a_k}{b_k}$$

exists, and $\rho \neq 0$, then either both series converge or both series diverge.

2. Use the Limit Comparison Test to determine whether the given infinite series converges or diverges.

(a) $\sum_{k=1}^{\infty} \frac{k+3}{2k^2+1}$

(b) $\sum_{k=1}^{\infty} \frac{k+\sqrt{k}}{k+k^3}$

(c) $\sum_{k=1}^{\infty} \frac{k^2-4}{k^3+k+5}$

(d) $\sum_{k=1}^{\infty} \frac{2k+2}{\sqrt{k^3+2}}$

Part V: Alternating Series

Absolute Convergence: The series $\sum a_k$ is said to **converge absolutely** if the series of absolute values, $\sum |a_k|$, converges.

1. (a) Show that the following series converges absolutely:

$$\sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{3k^2 + 1}$$

- (b) Show that the following series does not converge absolutely:

$$\sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{3k + 1}$$

Part V: Alternating Series (continued)

Absolute Convergence implies Convergence! If the series $\sum |a_k|$ converges then so does $\sum a_k$. That is, every absolutely convergent series converges.

2. (a) Show that the series

$$\sum_{k=1}^{\infty} \frac{(-1)^k k^2}{(k+1)!}$$

converges, by applying the Ratio Test to the series

$$\sum_{k=1}^{\infty} \left| \frac{(-1)^k k^2}{(k+1)!} \right|.$$

- (b) Demonstrate that the series

$$\sum_{k=1}^{\infty} \frac{\cos(\pi k)}{2^k}$$

converges, by showing that it converges absolutely.

Part V: Alternating Series (continued)

Alternating Series: An **alternating series** is a series of the form

$$\sum_{k=0}^{\infty} (-1)^k a_k = a_0 - a_1 + a_2 - a_3 + a_4 - \dots$$

where $a_k > 0$ for each $k = 0, 1, 2, 3, \dots$

The following theorem indicates the conditions under which an alternating series converges.

Alternating Series Test: Let $a_k > 0$ for each $k = 0, 1, 2, 3, \dots$. The alternating series

$$\sum_{k=0}^{\infty} (-1)^k a_k$$

converges if both the following hold:

- (i) The sequence $\{a_k\}_{k=0}^{\infty}$ is decreasing, and
- (ii) $\lim_{k \rightarrow \infty} a_k = 0$.

3. (a) Show that the series

$$\sum_{k=1}^{\infty} \frac{(-1)^{k+1} k}{k^2 + 1}$$

does not converge absolutely, by applying the Integral Test to the series

$$\sum_{k=1}^{\infty} \left| \frac{(-1)^{k+1} k}{k^2 + 1} \right|.$$

- (b) Demonstrate that the series

$$\sum_{k=1}^{\infty} \frac{(-1)^{k+1} k}{k^2 + 1}$$

converges, by using the Alternating Series Test.

Part V: Alternating Series (continued)

Conditional Convergence: The series $\sum a_k$ **converges conditionally** if $\sum a_k$ converges and $\sum |a_k|$ diverges.

4. Determine whether the series converges absolutely, converges conditionally, or diverges.

(a) $\sum_{k=1}^{\infty} \frac{(-1)^k}{2k+1}$

(b) $\sum_{k=1}^{\infty} \frac{(-1)^{k+1}k^2}{(k+2)!}$

(c) $\sum_{k=1}^{\infty} \frac{(-1)^k\sqrt{k}}{k^2+5}$

(d) $\sum_{k=1}^{\infty} \frac{(-1)^{k+1}k!}{6^k}$

Part VI: Ratio and Root Tests

The Ratio Test: Let $a_k > 0$ for all $k = 1, 2, \dots$ and let

$$\rho = \lim_{k \rightarrow \infty} \frac{a_{k+1}}{a_k}.$$

Then, provided the limit exists,

- (i) If $\rho < 1$, the series $\sum_{k=1}^{\infty} a_k$ converges.
- (ii) If $\rho > 1$, the series $\sum_{k=1}^{\infty} a_k$ diverges.
- (iii) If $\rho = 1$, no conclusion may be drawn.

1. Use the Ratio Test to determine whether the given infinite series converges or diverges.

(a) $\sum_{k=1}^{\infty} \frac{k^2}{3^k}$

(b) $\sum_{k=1}^{\infty} \frac{k^2}{(k+1)!}$

(c) $\sum_{k=1}^{\infty} \frac{k!}{k^k}$

Part VI: Ratio and Root Tests (continued)

The Root Test: Let $a_k > 0$ for all $k = 1, 2, \dots$ and let

$$\rho = \lim_{k \rightarrow \infty} \sqrt[k]{a_k}.$$

Then, provided the limit exists,

- (i) If $\rho < 1$, the series $\sum_{k=1}^{\infty} a_k$ converges.
- (ii) If $\rho > 1$, the series $\sum_{k=1}^{\infty} a_k$ diverges.
- (iii) If $\rho = 1$, no conclusion may be drawn.

2. Use the Root Test to determine whether the given infinite series converges or diverges.

(a) $\sum_{k=1}^{\infty} \frac{e^k}{k^k}$

(b) $\sum_{k=1}^{\infty} \left(\frac{k}{2k+1} \right)^k$

(c) $\sum_{k=1}^{\infty} \frac{k^3 2^{k+3}}{2^{2k}}$

Part VI: Ratio and Root Tests (continued)

3. Find those numbers $x > 0$ for which the given series converges.

(a) $\sum_{k=1}^{\infty} \frac{x^{2k}}{2^k}$

(b) $\sum_{k=1}^{\infty} \frac{x^k}{k!}$

End of Homework 5